YUZHONG CHEN

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EDUCATION

➢ **University of Liverpool (UoL)** **Liverpool, UK**
BSc Hons in Computer Science Sep. 2022 - May. 2026

- GPA: 3.88/4.0 | Major GPA: 4.0/4.0 | Rank: 1/214
- **Award:** Dean's List, Computer Science Departmental Prize, XJTLU Excellence Scholarship (£7,000 per year)
- **Fellowship:** EPSRC-funded Research Assistant, UoL, 2025 summer; Honorary Research Assistant, UoL, 2026-2027
- **Test Scores:** TOEFL 109 (R28, L29, S28, W24), GRE 329 (V159, Q170, AW3.5)

PAPERS

➢ [1] **Y. Chen**, Y. Qu, and H. Pu, "An analysis of attention mechanisms and its variance in transformer," *Applied and Computational Engineering*, vol. 47, no. 1, pp. 164-176, Mar. 2024, doi: 10.54254/2755-2721/47/20241291. (first author)

➢ [2] **Y. Chen**, T. Tan, A. Antala, S. Salaria, T. Saha, and Q. Tang, "Sequence Matters: A Controlled Study of Modality Injection Order in Multimodal LLMs," submitted to *ICLR 2027*, under review. (first author)

➢ [3] **Y. Chen**, D. Bollegala, "RG-TAIL: Reliability-Gated Trust-Aware Imagination Learning under Limited Budgets," submitted to *ICLR 2027*, under review. (first author)

➢ [4] **Y. Chen**, D. Bollegala, "Mixture of Encoders: Efficient Instance-Level Routing for Text Representation Learning," submitted to *ACL 2027*, under review. (first author)

RESEARCH PROJECTS

➢ **Mixture of Encoders: Efficient Instance-Level Routing** **Shenzhen, China**
Supervisor: Prof. Danushka Bollegala, Head of Machine Learning, UoL Jun. - Oct. 2026

- Proposed Mixture of Encoder (MoEn), trained a teacher router on expert embeddings, and distilled a student router that dynamically select the most appropriate expert encoder for a given text instance based on an anchor encoder embedding
- Evaluated MoEn against the strongest task-wise best fixed encoders baseline, demonstrated superior performance on both retrieval and classification task groups, proved the complementarity and effectiveness of the instance-level routing
- Conducted extensive analyses, proved that MoEn remains robust across different anchor encoders, while the optional top-k voting provides a flexible mechanism for trading additional encoder computation for improved performance

➢ **RG-TAIL: Reliability-Gated Trust-Aware Imagination Learning** **Shanghai, China**
Self-proposed Independent Research Apr. - Aug. 2026

- Conducted a large-scale empirical study on synthetic-data utilization in model-based reinforcement learning, revealing that uncertainty estimates do not necessarily reflect the long-term utility of synthetic transitions for policy optimization
- Proposed RG-TAIL, a framework that estimates transition-level trustworthiness from held-out real-world prediction errors, and introduces a reliability-gating mechanism based on world-model agreement and rollout consistency to adaptively determine whether synthetic experience should be utilized during downstream policy optimization
- Evaluated against external SOTA model-based RL methods (MBPO, TD-MPC2, DreamerV3), demonstrating superior average performance and task success rates under both extremely low- or high-budget settings (5k, 10k, 500k)

➢ **MERS: Modular Expert Routing System for Large Language Models** **Liverpool, UK**
Supervisor: Prof. Boris Konev, Dean of the School of Computer Science and Informatics, UoL Sep. 2025 - May. 2026

- Designed and implemented MERS, a modular architecture built from multiple domain-specialized expert models coordinated by a classifier-based routing mechanism; initialized two baselines: unified (single LLM fine-tuned on full corpus) and modular (per-domain experts along with classifier routing) on 10 disciplinary subsets from MMLU
- Evaluated MERS against unified baseline, demonstrating superior performance on datasets within the predefined expert pool, while losing cross-domain ability on 10 unseen MMLU subsets, along with increased inference latency and costs

➢ **Sequence Matters: A Controlled Study of Modality Injection Order in Multimodal LLMs** **Liverpool, UK**
Supervisors: Asst. Prof. Tulika Saha, UoL and *Asst. Prof. Qiyi Tang,* UoL Oct. 2024 - Jan. 2026

- Proposed MOCA-MLLM, a modality order-aware controlled architecture for MLLMs, with a unified preprocessing pipeline for twelve multimodal benchmark datasets (MELD, CMU-MOSI, HateMM, RLTD, etc.), enabling order-wise injection of text, vision, and audio embeddings at predefined decoder layers for large-scale, controlled ablation studies
- Conducted extensive fine-tuning experiments across four MLLMs, six downstream tasks, and six injection sequences, revealing task-dependent preferences: affective recognition tasks benefit from early visual/acoustic cues, whereas semantic-grounded tasks consistently favor early access to text, with text-middle configurations as the most suboptimal
- Demonstrated that carefully designed sequential modality injection consistently outperforms All-at-Input baselines, showing that increased processing depth alone does not guarantee improved performance

➢ **An Analysis of Attention Mechanisms and Its Variance in Transformer** **Beijing, China**

Supervisor: Assoc. Prof. Zengchang Qin, Beihang University Jun. - Sep. 2023

- Implemented and evaluated a suite of ML algorithms (DT, KNN, NB) and neural architectures including CNNs for license plate and traffic sign recognition, and RNNs (LSTM, BiLSTM) for sentiment and emotion classification
- Investigated Transformer variants, focusing on linearization, sparsification, multi-head self-attention and low-rank decomposition; conducted ablation studies to verify the performance improvement and highlight the trade-offs between scalability, interpretability and computational performance in Transformer-based architectures

PROFESSIONAL EXPERIENCE

➢ **UBTech Robotics Research Institute** **Shenzhen, China**

Embodied Foundation Model Department Sep. 2026 - Present

- Optimized the alignment of an in-house VLA foundation model built upon a pretrained VLM, by systematically adjusting the mixture of egocentric, real-world robotic, and simulation data to enhance cross-domain generalization
- Investigated Recursive Self-Improvement (RSI) techniques for embodied agents, leveraging Jev as a high-level action decision-maker; benchmarked against SOTA models on ManiSkill environments and documented a technical report

➢ **Shanghai Institute of Computing Technology** **Shanghai, China**

Research & Development Center Jul. - Sep. 2024

- Engineered and deployed a client-facing multimodal chatbot interface using Gradio with streaming inference and asynchronous I/O; supported text, image, file, and table inputs to demonstrate real-time reasoning and model capabilities
- Constructed a domain-specific agricultural knowledge base with LangChain and Word2Vec; optimized InternLM's RAG pipeline by integrating curated Q&A datasets and custom retrievers, improving top-k retrieval precision by 18%
- Developed a multi-turn real-time weather assistant agent, combining OpenWeatherMap API, prompt engineering and dynamic state tracking; conducted ablation studies on prompt fidelity, hallucination control and dialogue continuity

COURSEWORK PROJECTS

➢ **AI-Powered Recruitment Management System Development** Jan. - May. 2025

- Built a multi-user recruitment platform with candidate, recruiter, and admin modules; integrated OpenAI API and RAG for intelligent resume-job matching, and deployed backend via Flask and Alibaba Cloud with real-time chat functionality

➢ **Simulation and Optimization of Leader-Election Algorithms** Jan. - Mar. 2025

- Implemented distributed algorithms (LCR, HS) in Java; verified theoretical complexity bounds (O(n), O(n log n)) and reduced communication cost by up to 60% through algorithmic optimization and large-scale network simulations

➢ **Adversarial Attack and Defense in Neural Networks** Sep. - Dec. 2024

- Implemented framework (FGSM, PGD, adversarial training) on CNN baseline, ranking top 5% in robustness evaluation

➢ **Python-Based Data Visualization and Feature Analysis** Mar. - May. 2024

- Built a unified Python ML pipeline integrating feature extraction, dimensionality reduction (PCA, t-SNE, LDA), and both supervised and unsupervised learning; optimized classifier ensemble achieving top 5% performance among peers

ADDITIONAL

- **Programming Languages:** Python (4 yrs), Java (3 yrs), SQL (3 yrs), C (2 yrs), JavaScript (1 yr), HTML (1 yr)
- **Frameworks & Tools:** GitHub, GitLab, Hugging Face, Linux, ROS2, MySQL, Jupyter, MATLAB, Overleaf